

Accelerating Electric Vehicle Adoption

A County-Level Predictive Framework for Identifying High-Impact Regions

Springboard Capstone III Project Report

Sailee Gawande

Executive Summary

Electric vehicle (EV) adoption is growing quickly across the United States but unevenly across regions - even within the same state. This project develops a data-driven framework to quantify EV adoption at the county level, identify the socioeconomic and infrastructure factors that drive it, and segment regions into actionable investment tiers, using Washington State as a case study.

Two complementary models were built on a merged 2017–2025 county-year dataset spanning 39 Washington counties: a Ridge regression model to quantify drivers of EV adoption rate, and a K-Means clustering model to segment counties by adoption profile. The Ridge model explained approximately 96% of the variance in (transformed) adoption rate on a held-out 2024–2025 test set, with prior-year adoption momentum, median income, and charging infrastructure density emerging as the strongest predictors, in that order. The clustering model identified three distinct county segments — a high-income/high-adoption group, a low to mid income/infrastructure-supported group, and a single rural "charging desert" outlier (Garfield County), each requiring a different policy strategy.

These findings support a three-phase infrastructure and policy prioritization matrix:

1. Publicly funded foundational charging for the rural outlier county Garfield,
2. Demand-side subsidies and infrastructure maintenance for mid-income/infrastructure-rich counties, and
3. Reliance on private capital in high-income/high-adoption counties, with limited public investment reserved for underserved pockets.

1. Problem Identification & Context

Electric Vehicle adoption has grown significantly in the USA over the past decade, but adoption is very uneven across regions — some counties see rapid adoption while others lag behind despite being in the same state. This variation suggests that socioeconomic, technological, and policy factors interact to drive regional adoption, and this project explores those interactions.

Conventional (fossil-fuel) transportation contributed roughly 28% of U.S. greenhouse gas emissions in 2022 (EPA), making accelerated EV adoption important for emissions and sustainability goals. However, EV infrastructure — charging networks, incentives, and supply chain investment — is expensive and must be strategically targeted; without data-driven prioritization, investment risks being misallocated to low-impact regions.

This project's objective is to develop a data-driven framework to quantify EV adoption at the county level and identify its key drivers, using Washington State as a case study. EV adoption rate is constructed as the proportion of electric vehicle registrations to total passenger vehicle registrations, at county/year granularity.

1.1 Criteria for Success

- A Ridge regression model that achieves meaningful explanatory power and identifies significant features driving regional EV adoption.
- Identification of interpretable, stable EV adoption segments (clusters) to support clear policy strategy.
- Actionable recommendations for EV infrastructure investment and policy targeting.

1.2 Scope

In scope:

1. Feature engineering and target variable construction,
2. Regional EV adoption analysis
3. Predictive modeling of adoption potential, county segmentation
4. Business recommendation framework.

Out of scope:

1. Individual consumer-level behavior modeling (not available in the data) and
2. Charging network route-level optimization.

1.3 Stakeholders

- Electric vehicle manufacturers
- Government and transportation policymakers (incentive allocation)
- EV charging infrastructure providers
- Utility companies
- Sustainability investors

2. Data Sources

All data sources are Washington State public data, aggregated to county/year granularity for 2017–2025:

Source	Description
Electric Vehicle Population Data (Data.WA.gov)	Washington DOL registered EV records: make, model, type, electric range, CAFV eligibility

Source	Description
Vehicle Registrations by Class and County (Data.WA.gov)	All passenger vehicle registrations; used as the adoption-rate denominator
Median Household Income Estimates (OFM)	County-level median household income
Population Estimates (OFM) Population Archive	County-level population
EV Charging Stations Data (WSDOT)	Public EV charging station locations and counts

The median income, population, and charging station datasets were pre-merged into a single supporting file prior to the data wrangling step.

3. Methodology

This project followed the standard data science method: problem identification, data wrangling, exploratory data analysis, preprocessing and training/test set development, modeling, and documentation.

3.1 Data Wrangling

Three data sources — EV registrations, all-vehicle registrations, and socioeconomic/infrastructure data — were cleaned, standardized, and merged to the county-year grain (39 Washington counties, 2017–2025).

- Filtered EV data to Washington State only; dropped fields not needed for county-level analysis (Legislative District, Vehicle Location, 2020 Census Tract).
- Imputed a small number of missing Electric Range values using make/model medians, with one external lookup applied for a model with no comparable in-dataset records.
- Aggregated EV records to county-year level: BEV/PHEV mix, CAFV eligibility rate, and EV counts by model year.
- Aggregated all-vehicle registrations to county-year totals and EV-only subtotals, then merged with the EV aggregate.
- Merged in the socioeconomic/infrastructure support dataset (population, median income, charging stations) by county and year.
- **Target variable — EV Adoption Rate:** (registered electric vehicles ÷ total registered passenger vehicles,) x 100, at county/year level.

Additional engineered features included EV growth rate, EV adoption lag (constructed using strictly prior-year values to prevent temporal leakage), a high-adoption flag, adoption segment tiers, income per vehicle, and EV charging density (charging stations per capita).

During validation, average BEV range was found to be implausibly low starting around 2021; further research confirmed that Washington State stopped tracking BEV electric range in its official data after 2020. The range-related fields (bev_avg_range, phev_avg_range,

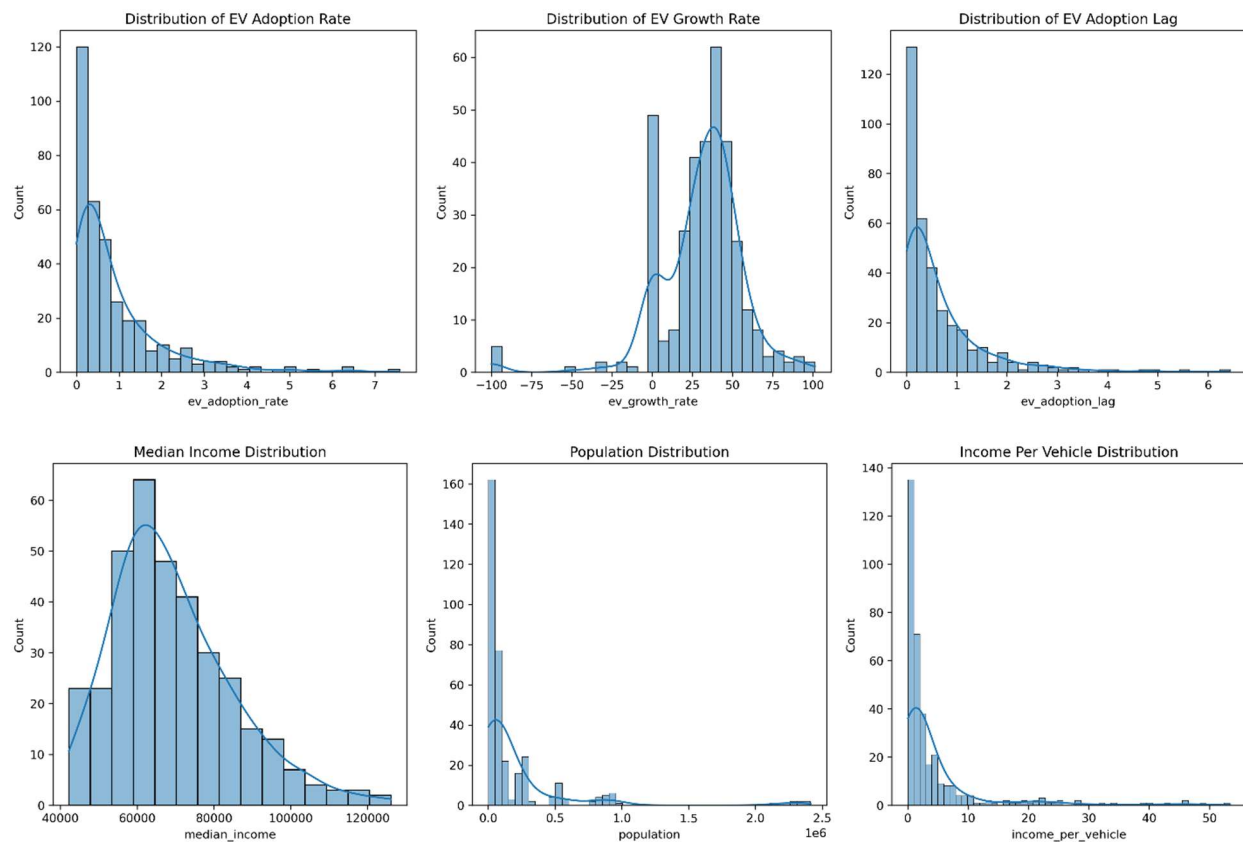
range_missing_rate) were dropped as unusable for modeling — a data availability limitation rather than a modeling choice.

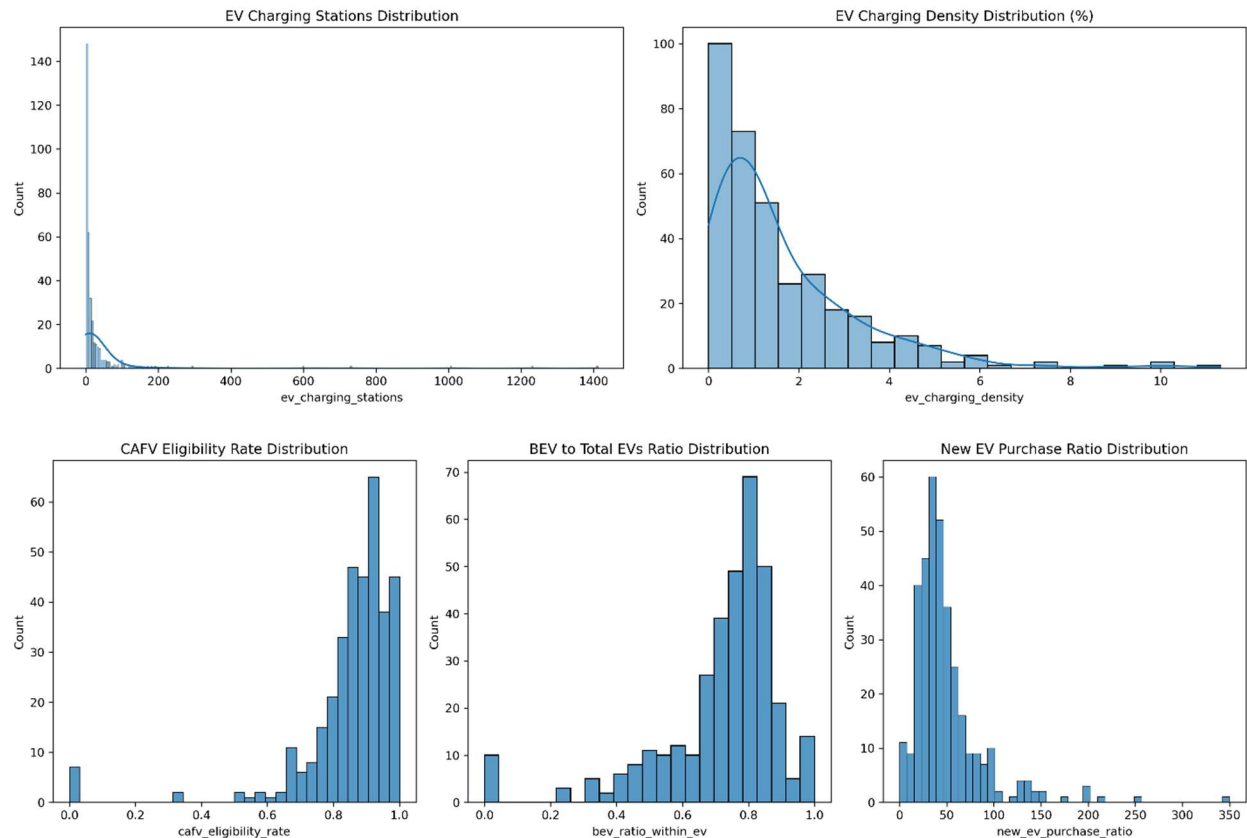
3.2 Exploratory Data Analysis

EDA focused on understanding feature and target distributions, feature relationships, time trends, and model readiness.

3.2.1 Distributions

EV adoption rate, EV adoption lag, population, and EV charging density/stations were all right-skewed, indicating a need for transformation prior to modeling. CAFV eligibility rate and BEV share (from all EVs) were left-skewed, reflecting the dominance of high-eligibility, BEV share (from amongst all EVs)-heavy counties.



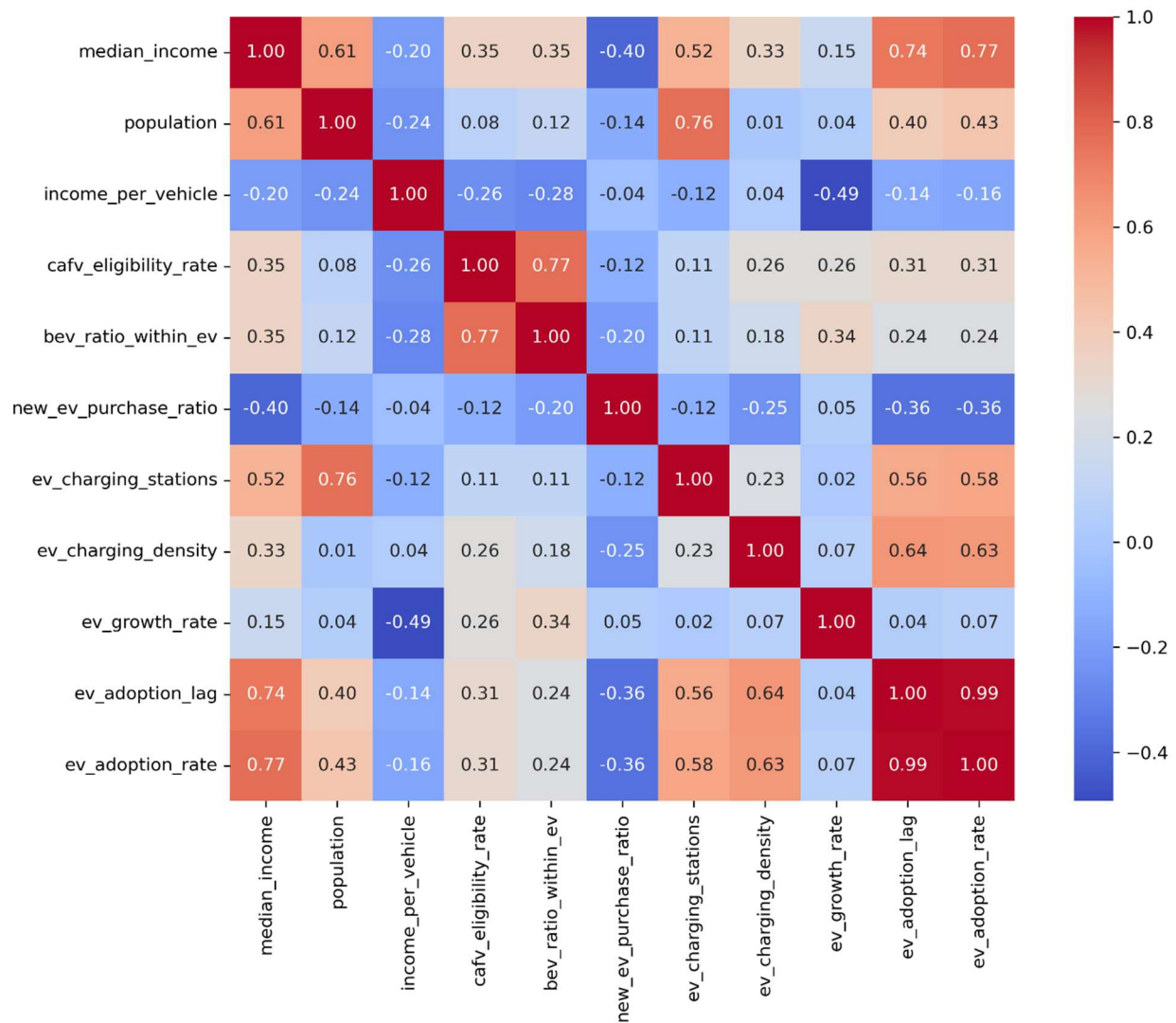


3.2.2 Bivariate Relationships with Adoption Rate

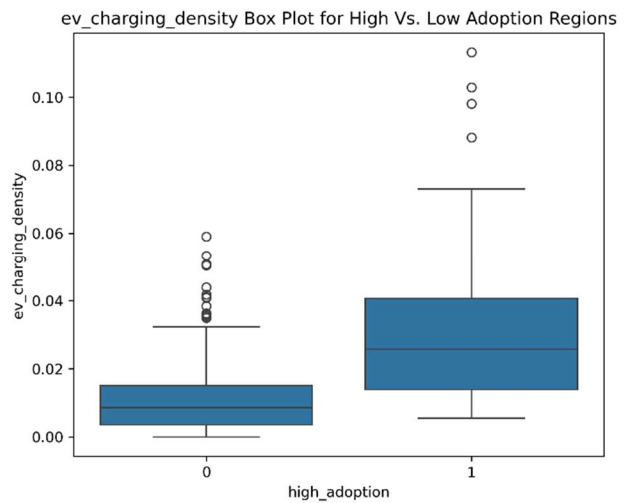
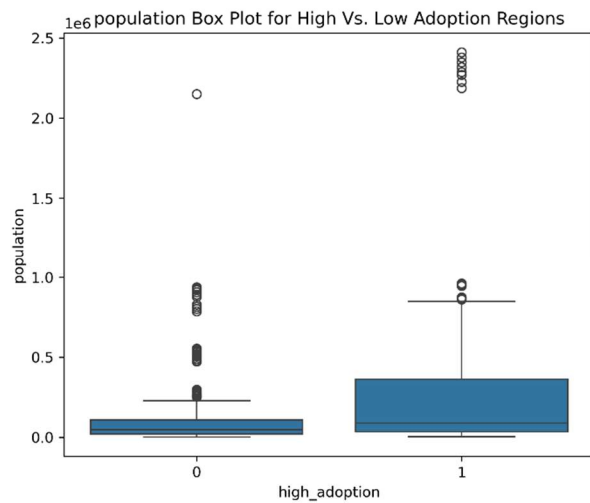
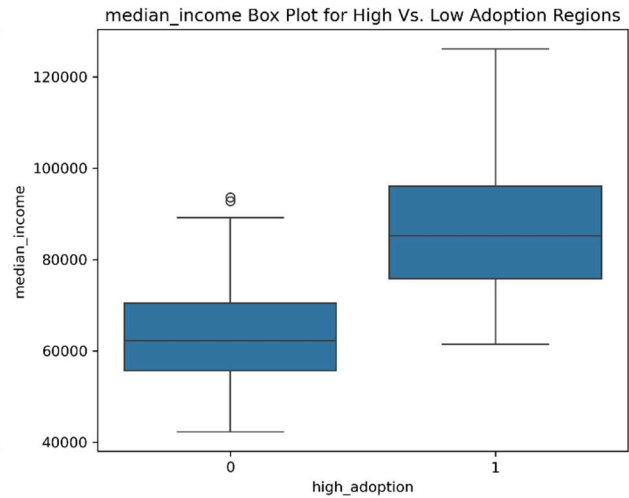
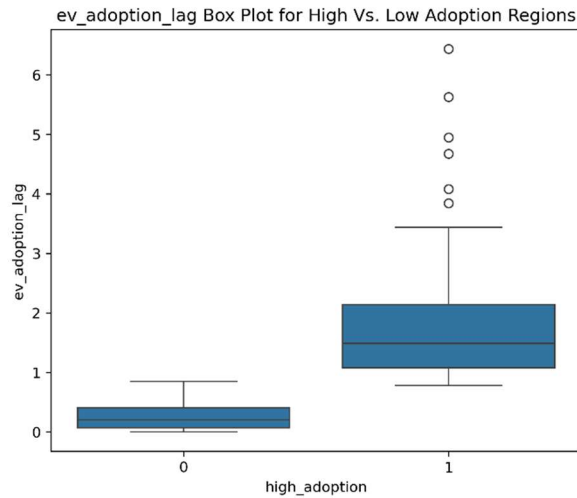
Feature	Correlation with EV Adoption Rate
ev_adoption_lag	0.99
median_income	0.77
ev_charging_density	0.63
ev_charging_stations	0.58
population	0.43
cafv_eligibility_rate	0.31
bev_ratio_within_ev	0.24
ev_growth_rate	0.07
income_per_vehicle	-0.16 (distorted by rural outliers)
new_ev_purchase_ratio	-0.36 (distorted by reporting lag)

EV Adoption Lag, Median Income & EV Charging Density features seem to have the highest correlation with EV Adoption.

Income per vehicle and new EV purchase ratio both showed extreme outlier-driven distortion (small rural vehicle counts and end-of-timeline registration reporting lag, respectively) and were excluded from modeling rather than corrected, since the anomalies reflected data artifacts rather than genuine signal.

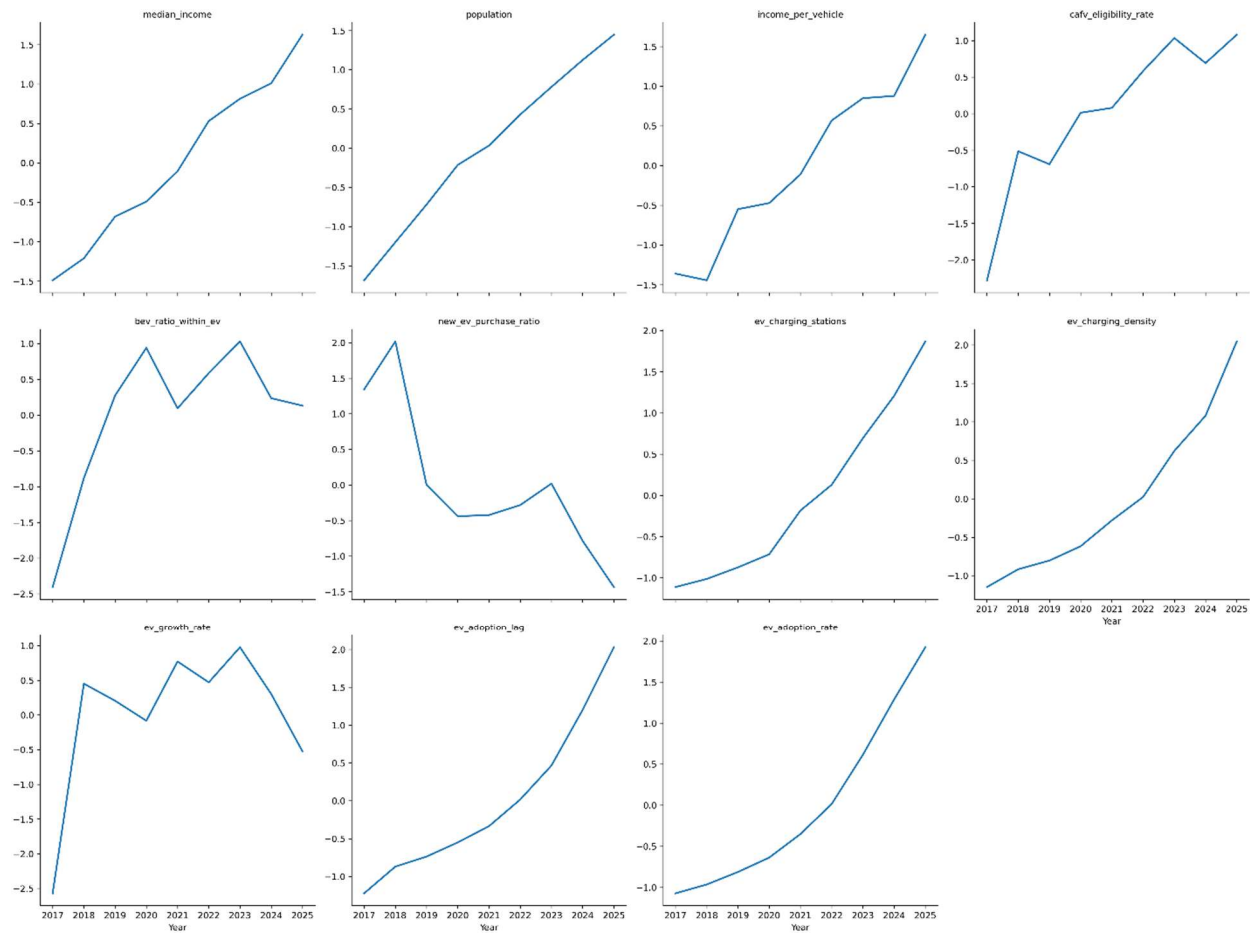


The boxplots reveal a clear socio-economic divide where high-adoption regions are clearly separated by higher median incomes, charging station densities, and historical adoption momentum, while highlighting extreme population outliers and the heavy visual redundancy of the vehicle-mix features such as `cafv_eligibility_rate` & `bev_ratio_within_ev` (showing compressed, high-positioned medians for high-adoption zones).



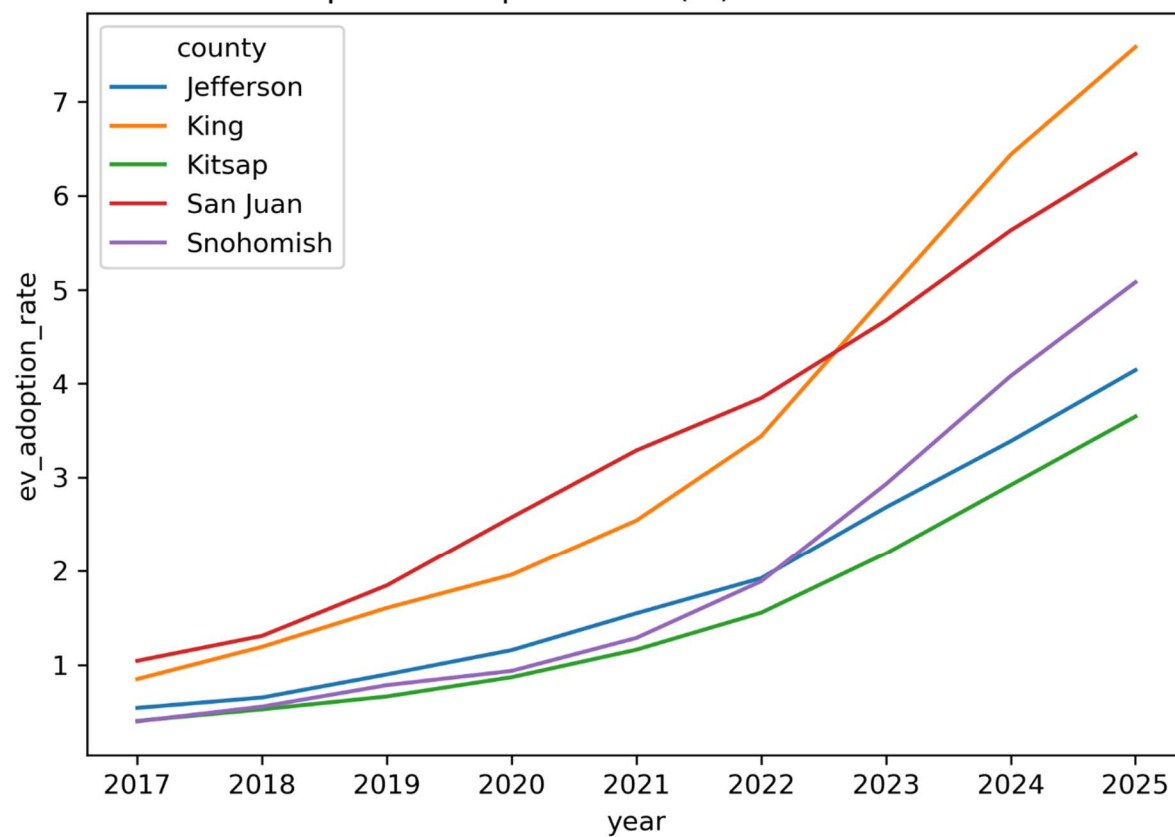
3.2.3 Time Trends

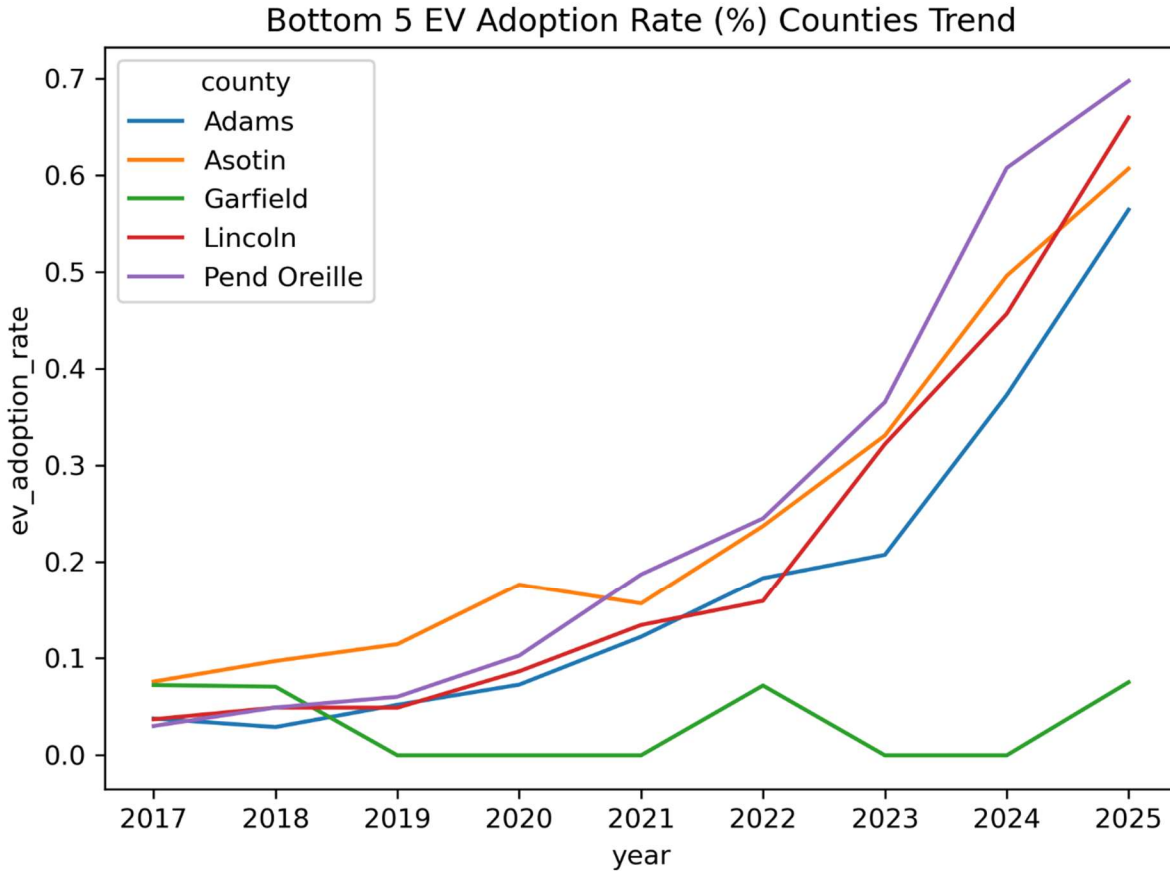
Socioeconomic indicators (income, population) grew steadily from 2017–2025. EV adoption rate, adoption lag, and charging infrastructure all showed accelerating, near-exponential growth over the same period. CAFV eligibility and EV technology mix showed early momentum followed by stabilization, consistent with a maturing market.



The top 5 vs. bottom 5 counties EV Adoption trends (below) show a huge divide where high ev adoption counties have achieved an exponential momentum while low adoption counties are seeing 0 to extremely low adoption potentially due to low population, lack of income, infrastructure and policies.

Top 5 EV Adoption Rate (%) Counties Trend

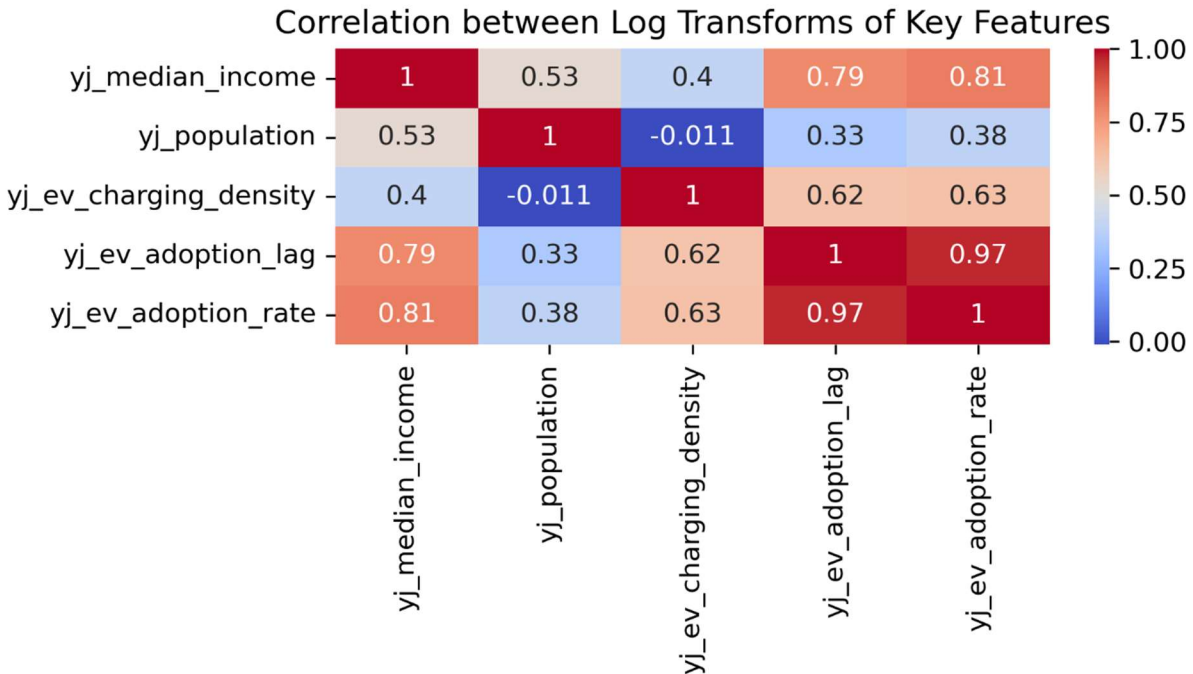




3.2.4 Feature Transformation & Multicollinearity

Log transformation reduced but did not fully normalize skew in several key features. A Yeo–Johnson power transform was applied instead to population, median income, EV charging density, EV adoption lag, and the target variable, producing more model-friendly (though still non-normal, block-shaped) distributions. Because Ridge regression's core requirement is normally distributed residuals — not normally distributed inputs — and Ridge's L2 penalty stabilizes coefficients under non-normal feature distributions, this was judged acceptable. A variance inflation factor (VIF) check confirmed low multicollinearity across the retained feature set (all VIF values below 4.3).

Weakly correlated or anomaly-prone features (`income_per_vehicle`, `new_ev_purchase_ratio`, `bev_ratio_within_ev`, `cafv_eligibility_rate`, `ev_charging_stations`, `ev_growth_rate`) were dropped from the modeling dataset, leaving a compact, low-collinearity feature set.



3.3 Preprocessing & Training Set Development

Two feature sets were finalized for the two modeling approaches:

- **Ridge Regression features:** yj_population, yj_median_income, yj_ev_adoption_lag, yj_ev_charging_density, has_charging_infrastructure, lag_missing_flag
- **K-Means features:** yj_population, yj_median_income, yj_ev_charging_density, has_charging_infrastructure

County was intentionally excluded as a model feature — for regression, to keep coefficients generalizable rather than county-memorized; for clustering, to avoid geography-driven clusters that would defeat the purpose of behavioral segmentation. County and year were retained as reference columns for interpretation.

Because the dataset is a county-year panel / time series, a time-aware split was used: training data covers 2017–2023 and test data covers 2024–2025 (an approximately 80/20 split). A random split was avoided to prevent future information from leaking into training. Numeric features were standardized with a StandardScaler fit on the training data only (for the regression set); no train/test split was applied for the unsupervised K-Means model, which used the full dataset with the adoption rate retained purely as a post-hoc interpretive label.

3.4 Modeling

3.4.1 Ridge Regression

Ridge regression was selected over ordinary least squares because of the modest test set size (~78 rows) - Ridge's L2 regularization reduces overfitting risk while retaining all features (as

opposed to Lasso, which can zero out coefficients entirely). The regularization strength (alpha) was tuned via GridSearchCV using TimeSeriesSplit cross-validation (5 folds) to respect the temporal structure of the data and avoid leakage across folds.

3.4.2 K-Means Clustering

For county segmentation, K-Means was applied to the latest available year of standardized data for each of the 39 counties. The optimal number of clusters (k=3) was selected using the elbow method and confirmed by a peak in silhouette score.

4. Results

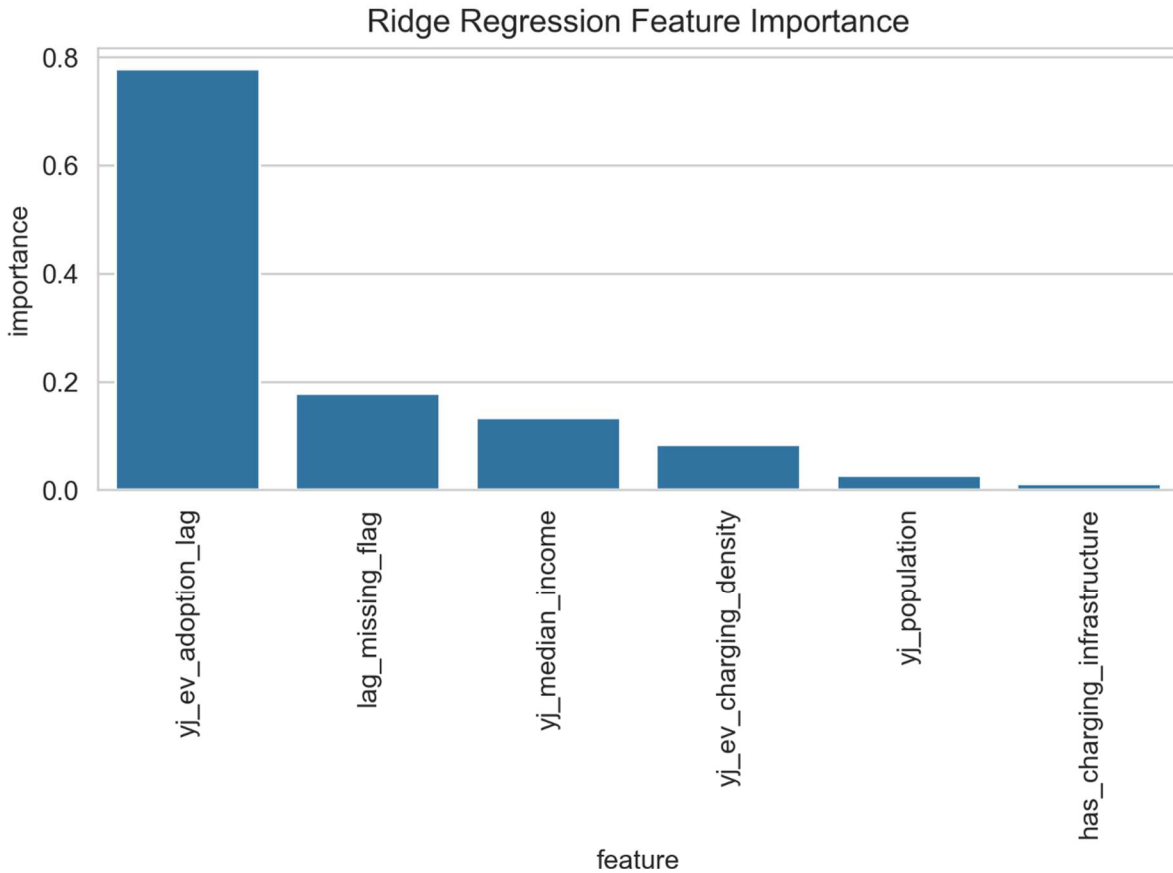
4.1 Ridge Regression: Adoption Drivers

Metric	Train (2017–2023)	Test (2024–2025)
R ²	≈ 0.95	≈ 0.96
RMSE	≈ 0.2	≈ 0.14
MAE	≈ 0.13	≈ 0.11

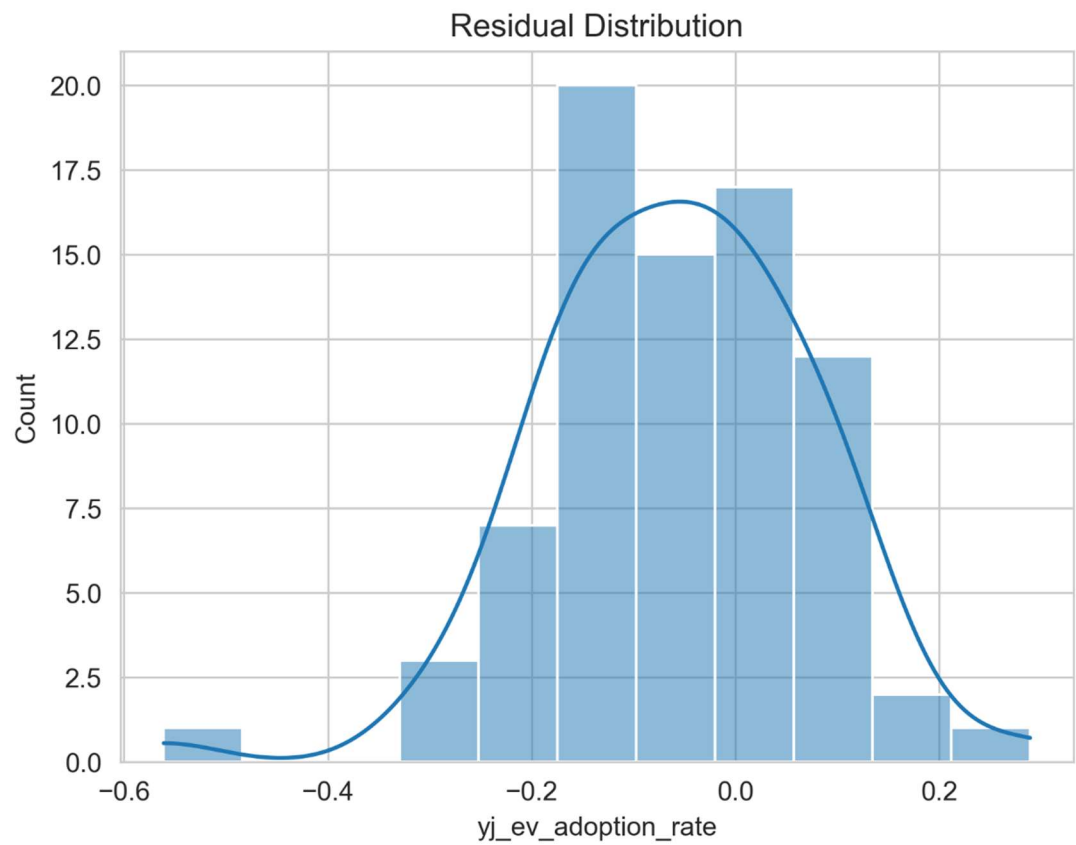
(Metrics computed on the Yeo–Johnson-transformed adoption rate. R² of ~0.96 on the held-out 2024–2025 test set indicates the model explains the large majority of variance in county-level adoption; RMSE and MAE are both low, indicating predictions are close to actual values.)

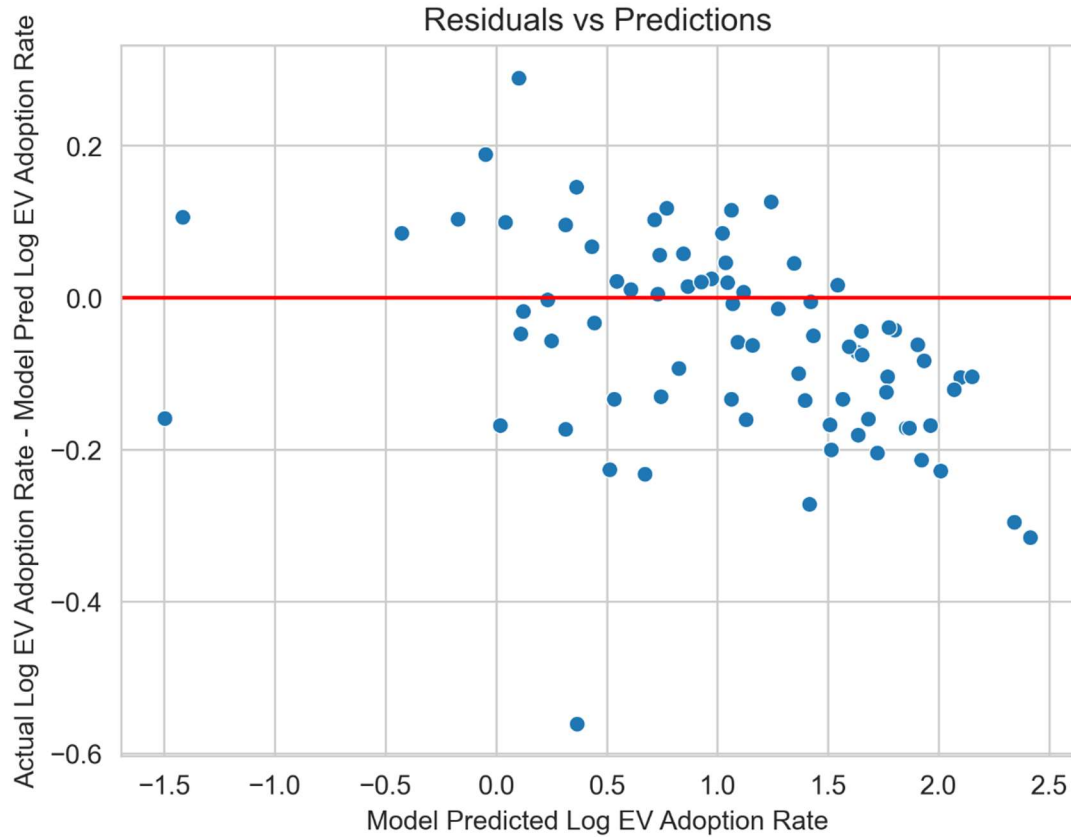
Feature importance, assessed via standardized Ridge coefficients, in descending order:

- EV adoption lag (prior-year adoption) — by far the dominant driver; historical adoption momentum strongly predicts future adoption.
- Median income — a strong secondary driver, slightly ahead of charging infrastructure, indicating infrastructure investment alone is unlikely to drive adoption without supportive socioeconomic conditions.
- EV charging density and has_charging_infrastructure — meaningful positive contributors.
- Population — a comparatively weak driver; higher population alone does not guarantee higher adoption rate.



Residual analysis showed errors roughly centered at zero and approximately normally distributed, satisfying the core linear regression assumption on residuals. A mild tendency to over-predict at higher adoption values and under-predict at lower values was observed, suggesting some residual non-linearity, but overall, the model captures the dominant signal well and provides a decent fit.



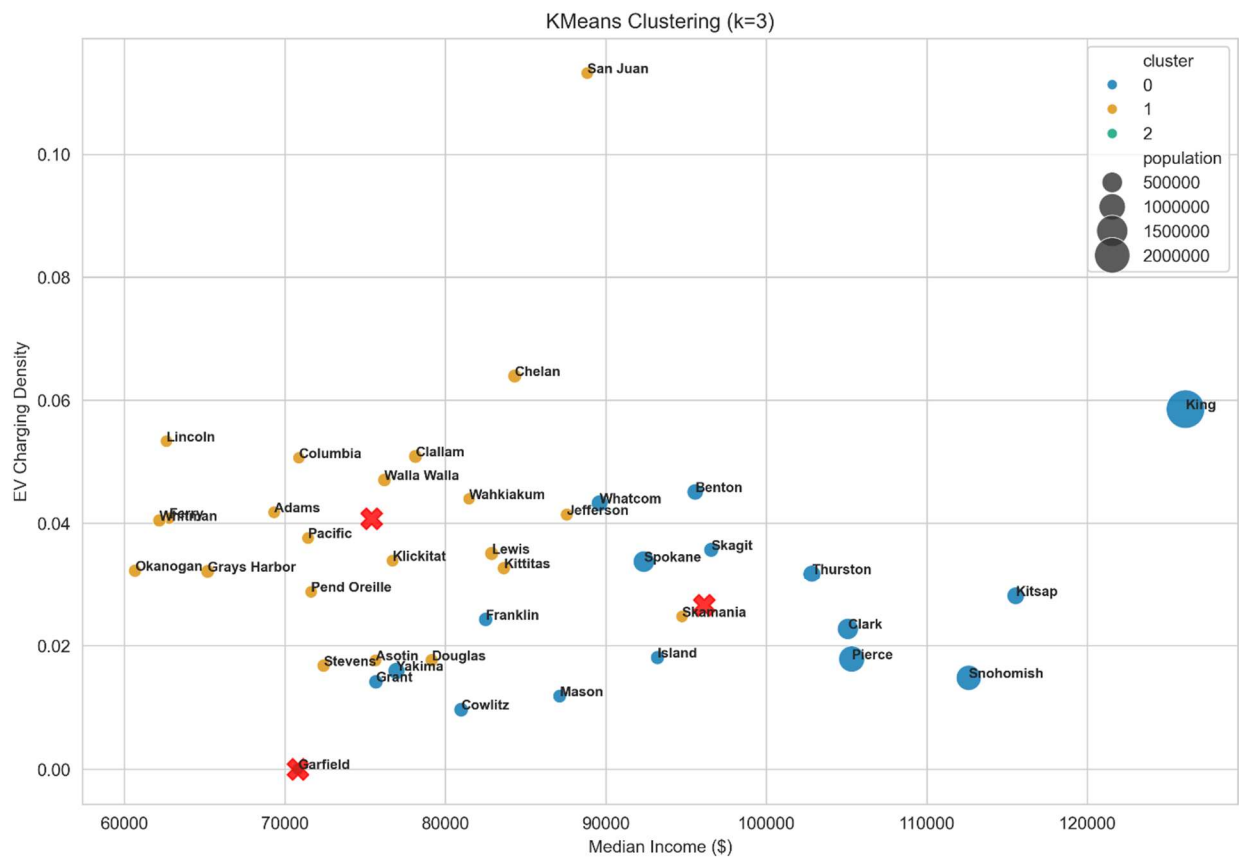


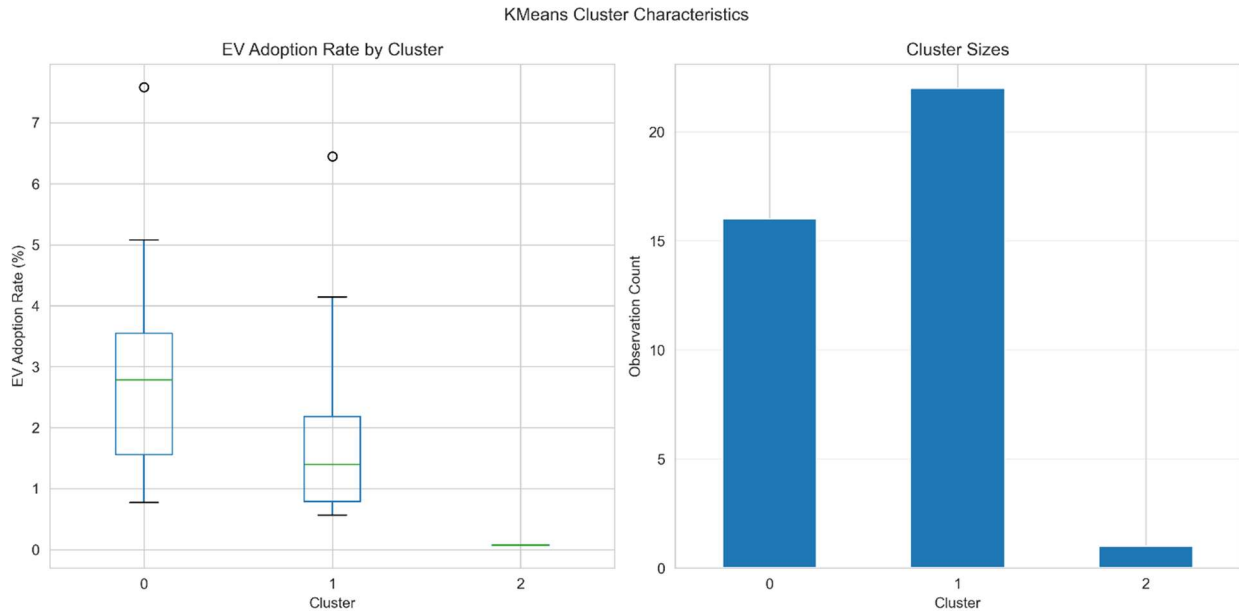
4.2 K-Means Clustering: County Segmentation

Cluster	Counties	Avg. Adoption Rate	Avg. Income	Avg. Charging Density	Profile
0 — High Adoption	16 (41%)	~3.0%	~\$96K	~0.03	High income, high population; adoption driven by purchasing power / likely home charging
1 — Medium Adoption	22 (56%)	~1.6%	~\$60K–95K	~0.04	Lower income, lower population, but strong public charging density

Cluster	Counties	Avg. Adoption Rate	Avg. Income	Avg. Charging Density	Profile
2 — Rural Isolation Outlier	1 (Garfield)	~0.08%	~\$71K	~0.00	Small, rural, zero public charging infrastructure — a structural charging desert

Cluster 0 counties (e.g., King, Snohomish, Clark) lead the state in adoption and represent self-sustaining markets where private capital can drive further infrastructure buildout. Cluster 1 counties (e.g., Whitman, Adams, Okanogan, Jefferson) have already solved much of the infrastructure gap and are primarily limited by vehicle affordability. Cluster 2 (Garfield County) is an outlier with comparable income to parts of Cluster 1 but zero charging infrastructure and negligible population, resulting in adoption trapped near zero — momentum cannot start without a foundational infrastructure investment.





4.3 Combined Insights

- **Income Elasticity Threshold:** Higher income triggers the adoption momentum loop, explaining Cluster 0's leading ~3% adoption rate — but K-Means shows high income is not strictly required if infrastructure exists (Cluster 1).
- **Infrastructure as an Equalizer:** Cluster 1, despite a lower income profile, sustains meaningful adoption through higher public charging density — infrastructure can partly substitute for income.
- **Structural Starvation Baseline:** Cluster 2 (Garfield) has income comparable to parts of Cluster 1 but due to zero infrastructure and low population, adoption remains near a structural floor. Momentum cannot begin without an infrastructure foundation.

5. Business Recommendations: Three-Phase Prioritization Matrix

Combining the regression-identified drivers with the cluster-based segments produces a deployment blueprint that matches funding mechanism and strategy to each group's specific constraints.

Phase	Priority	Target Counties	Core Strategy	Funding Mechanism
1 — Foundation	Critical / Immediate	Garfield	Seed baseline highway-corridor charging to eliminate the state's only total charging desert	100% public capital (NEVI / state clean energy grants)
2 — Activation	Demand Activation	Cluster 1 counties (e.g., Whitman, Adams, Okanogan, Lincoln, and	Vehicle purchase rebates and used-BEV subsidies to unlock pent-up demand; maintain	Point-of-sale rebates, used-BEV subsidies,

Phase	Priority	Target Counties	Core Strategy	Funding Mechanism
		other lower-income, infrastructure-rich counties)	and enforce DC fast-charger uptime reliability	public–private matching funds
3 — Autonomy	Sustaining Market	Cluster 0 counties (King, Snohomish, Clark, Pierce, and other high-income, high-population counties)	Rely on private commercial capital for destination charging; limit public spending to grid-capacity matching and select low-income pockets	Mostly private / commercial capital

6. Limitations & Assumptions

- Analysis is limited to Washington State; the framework is designed to be portable but has not been validated on other states.
- EV adoption rate is a constructed label (registered EVs ÷ total registered vehicles) rather than a directly reported metric.
- Electric range tracking for BEVs was discontinued in state data after 2020, so range-based features could not be used as predictors.
- New EV purchase ratio and income-per-vehicle showed clear data artifacts (registration reporting lag, rural outlier distortion) and were excluded rather than corrected.
- The K-Means model uses each county's most recent available year only, so it reflects a snapshot rather than a full adoption trajectory.
- Sample size is modest (39 counties across up to nine years), and one cluster consists of a single county — results should be interpreted directionally rather than as precise point estimates.
- This project models regional/aggregate adoption; it does not model individual consumer purchase behavior, which was not available in the source data.

7. Future Work

Several extensions would strengthen and broaden this framework:

- **Geographic generalization:** Validate the framework on additional states beyond Washington to test portability and identify which drivers are state-specific versus broadly generalizable.
- **Finer spatial granularity:** Move from county-level to ZIP-code or census-tract level analysis where data allows, to better target infrastructure within large or geographically diverse counties.

- **Charging infrastructure detail:** Differentiate charger types (Level 2 vs. DC fast) and station reliability/uptime, rather than treating all charging infrastructure as equivalent.
- **Time-series forecasting:** Extend beyond a static driver model to explicitly forecast county-level adoption rates several years forward (e.g., ARIMA, or panel-data models), supporting longer-horizon infrastructure planning.
- **Model refinement:** Test non-linear models (e.g., gradient boosting, random forest) to capture the mild residual non-linearity observed in the Ridge model, while weighing interpretability trade-offs against the current model's transparency.
- **Deployment:** Package the framework as an interactive dashboard (e.g., Tableau) so policymakers and infrastructure planners can explore county-level drivers and recommendations directly and refresh it as new registration data becomes available.

8. Conclusion

This project delivers a reusable, data-driven framework for identifying and prioritizing high-impact regions for EV infrastructure investment. The Ridge regression model quantifies which factors drive county-level adoption — prior adoption momentum, income, and infrastructure, in that order of importance — while the K-Means clustering model translates those drivers into three actionable policy segments. Together, they support a phased, funding-mechanism-specific deployment strategy that concentrates scarce public capital where it will have the greatest impact, while allowing private capital to scale adoption in self-sustaining markets.